Lab 1 - DAV

Name : Piyush Khokale

Section: A-A2

Roll no: 17

In [1]:
```
import pandas as pd
```

In [3]:
```
df_csv=pd.read_csv("student_data.csv")
print(df_csv.head())
```

```
   Stud_id      Stud_name Branch Section
0      101  Alice Johnson    CSE       A
1      102      Bob Smith    ECE       B
2      103  Charlie Davis     ME       C
3      104   David Miller     CE       D
4      105     Eve Wilson     IT       A
```

In [4]:
```
with open("The deep-sea.txt","r") as file:
    content1=file.read()
    print(content1)
```

The deep-sea abyssal zoneâ€”spanning depths from 3,000 to 6,000 meters below the ocean surfaceâ€”remains one of the most enigmatic, hostile, and endlessly fascinating biomes on Earth. In this realm of perpetual darkness, the sunâ€™s rays are completely absent, leaving the water at a near-freezing temperature just above zero degrees Celsius. The environmental pressure here is immense, crushing down with a force equivalent to an elephant standing on a postage stamp for every square inch of surface area. For centuries, scientists assumed this vast desert of abyssal plains was entirely devoid of life. However, modern oceanographic exploration has revealed a thriving, surreal ecosystem populated by organisms that have undergone radical evolutionary adaptations to survive. Without sunlight to drive photosynthesis, the food web relies heavily on "marine snow"â€”a continuous drift of organic detritus, dead plankton, and fecal matter sinking from the upper layers of the ocean. In pockets where tectonic plates shift, hydrothermal vents spew toxic, mineral-rich fluids into the water. Here, specialized bacteria utilize chemosynthesis rather than photosynthesis to convert these chemicals into energy, forming the foundation for dense communities of giant tube worms, blind shrimp, and pale crabs that flourish completely independent of solar energy.

The creatures that inhabit the abyssal zone exhibit bizarre anatomical features designed to maximize their chances of survival in the absolute dark. Bioluminescence is highly prevalent; many organisms generate their own cold light through complex chemical reactions to attract mates, lure unsuspecting prey, or startle potential predators. The deep-sea anglerfish, for instance, uses a glowing dorsal filament dangling like a fishing rod over its massive, needle-fanged jaws. Because food is incredibly scarce, many abyssal fish have evolved highly distensible stomachs and expandable jaws, allowing them to swallow prey much larger than themselves in a single meal. Others, like the tripod fish, use elongated pelvic and caudal fins to stand rigidly on the muddy seafloor, facing into the slow current to catch tiny crustaceans drifting by. Despite its isolation, this fragile ecosystem faces mounting modern threats from human activity, including climate-driven changes in marine snow production, plastic pollution filtering down from the surface, and the looming prospect of deep-sea mining for rare Earth minerals. Exploring and protecting these deep frontiers is crucial, as they not only hold the secrets to extreme adaptation and the origins of life on Earth but also regulate global carbon cycles in ways we are only beginning to fully comprehend.

In [6]:
```
df_txt = pd.read_csv("The deep-sea.txt", delimiter="\t") # or delimiter=","
print(df_txt.head())
```

```
  The deep-sea abyssal zone—spanning depths from 3,000 to 6,000 meters below
the ocean surface—remains one of the most enigmatic, hostile, and endlessly
fascinating biomes on Earth. In this realm of perpetual darkness, the sun’s
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just above zero degrees Celsius. The environmental pressure here is immense,
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and fecal matter sinking from the upper layers of the ocean. In pockets wher
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into the water. Here, specialized bacteria utilize chemosynthesis rather tha
n photosynthesis to convert these chemicals into energy, forming the foundat
ion for dense communities of giant tube worms, blind shrimp, and pale crabs
that flourish completely independent of solar energy.
0  The creatures that inhabit the abyssal zone ex...
```

In [7]:
```
df_txt = pd.read_csv("The deep-sea.txt", delimiter=",") # or delimiter="," 
print(df_txt.head())
```

    The deep-sea abyssal zone—spanning depths from 3   000 to
6 \
0 The creatures that inhabit the abyssal zone ex...   lure unsuspecting pre
y

  000 meters below the ocean surface—remains one of the most enigmatic \
0   or startle potential predators. The deep-sea ...

         hostile \
0   for instance

   and endlessly fascinating biomes on Earth. In this realm of perpetual dar
kness \
0   uses a glowing dorsal filament dangling like ...

               the sun’s rays are completely absent \
0   needle-fanged jaws. Because food is incredibl...

   leaving the water at a near-freezing temperature just above zero degrees
Celsius. The environmental pressure here is immense \
0   many abyssal fish have evolved highly distens...

   crushing down with a force equivalent to an elephant standing on a postag
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0   allowing them to swallow prey much larger tha...

   scientists assumed this vast desert of abyssal plains was entirely devoid
of life. However \
0   like the tripod fish

   modern oceanographic exploration has revealed a thriving \
0   use elongated pelvic and caudal fins to stand...

   surreal ecosystem populated by organisms that have undergone radical evol
utionary adaptations to survive. Without sunlight to drive photosynthesis \
0   facing into the slow current to catch tiny cr...

   the food web relies heavily on "marine snow"—a continuous drift of organi
c detritus \
0   this fragile ecosystem faces mounting modern ...

                                      dead plankton \
0   including climate-driven changes in marine sn...

   and fecal matter sinking from the upper layers of the ocean. In pockets w
here tectonic plates shift \
0   plastic pollution filtering down from the sur...

                       hydrothermal vents spew toxic \
0   and the looming prospect of deep-sea mining f...

            mineral-rich fluids into the water. Here \
0   as they not only hold the secrets to extreme ...

    specialized bacteria utilize chemosynthesis rather than photosynthesis t
o convert these chemicals into energy \

```
0                                                   NaN

    forming the foundation for dense communities of giant tube worms  \
0                                                   NaN

    blind shrimp  \
0            NaN

    and pale crabs that flourish completely independent of solar energy.
0                                                   NaN
```

In [8]:
```
import json

with open("people_data.json", "r") as file:
    data_json = json.load(file)
    print(data_json)
df_json = pd.DataFrame(data_json)
print(df_json.head())
```

```
[{'name': 'Elena Rostova', 'age': 28, 'country_of_origin': 'Russia'}, {'nam
e': "Liam O'Connor", 'age': 34, 'country_of_origin': 'Ireland'}, {'name': 'A
iko Tanaka', 'age': 22, 'country_of_origin': 'Japan'}, {'name': 'Mateo Silva
', 'age': 41, 'country_of_origin': 'Brazil'}, {'name': 'Fatima Al-Sayed', 'a
ge': 29, 'country_of_origin': 'Egypt'}, {'name': 'Kwame Mensah', 'age': 37,
'country_of_origin': 'Ghana'}, {'name': 'Chloe Dubois', 'age': 25, 'country_
of_origin': 'France'}, {'name': 'Arjun Patel', 'age': 31, 'country_of_origin
': 'India'}, {'name': 'Sofia Nielsen', 'age': 45, 'country_of_origin': 'Denm
ark'}, {'name': 'Lucas Miller', 'age': 53, 'country_of_origin': 'Canada'}]
              name  age country_of_origin
0    Elena Rostova   28            Russia
1    Liam O'Connor   34           Ireland
2      Aiko Tanaka   22             Japan
3      Mateo Silva   41            Brazil
4  Fatima Al-Sayed   29             Egypt
```

In [10]:
```
import xml.etree.ElementTree as ET

tree=ET.parse("people_data.xml")
root=tree.getroot()

for elem in root.iter():
    print(elem.tag, elem.text)
```

```
People

Person

Name Elena Rostova
Age 28
CountryOfOrigin Russia
Person

Name Liam O'Connor
Age 34
CountryOfOrigin Ireland
Person

Name Aiko Tanaka
Age 22
CountryOfOrigin Japan
Person

Name Mateo Silva
Age 41
CountryOfOrigin Brazil
Person

Name Fatima Al-Sayed
Age 29
CountryOfOrigin Egypt
Person

Name Kwame Mensah
Age 37
CountryOfOrigin Ghana
Person

Name Chloe Dubois
Age 25
CountryOfOrigin France
Person

Name Arjun Patel
Age 31
CountryOfOrigin India
Person

Name Sofia Nielsen
Age 45
CountryOfOrigin Denmark
Person

Name Lucas Miller
Age 53
CountryOfOrigin Canada
```

In [11]:
```python
import json

with open("people_data.json", "r") as file:
```

```
    data_json = json.load(file)
    print(data_json)
df_json = pd.DataFrame(data_json)
print(df_json.tail())
```

```
[{'name': 'Elena Rostova', 'age': 28, 'country_of_origin': 'Russia'}, {'nam
e': "Liam O'Connor", 'age': 34, 'country_of_origin': 'Ireland'}, {'name': 'A
iko Tanaka', 'age': 22, 'country_of_origin': 'Japan'}, {'name': 'Mateo Silva
', 'age': 41, 'country_of_origin': 'Brazil'}, {'name': 'Fatima Al-Sayed', 'a
ge': 29, 'country_of_origin': 'Egypt'}, {'name': 'Kwame Mensah', 'age': 37,
'country_of_origin': 'Ghana'}, {'name': 'Chloe Dubois', 'age': 25, 'country_
of_origin': 'France'}, {'name': 'Arjun Patel', 'age': 31, 'country_of_origin
': 'India'}, {'name': 'Sofia Nielsen', 'age': 45, 'country_of_origin': 'Denm
ark'}, {'name': 'Lucas Miller', 'age': 53, 'country_of_origin': 'Canada'}]
            name  age country_of_origin
5   Kwame Mensah   37            Ghana
6   Chloe Dubois   25           France
7    Arjun Patel   31            India
8  Sofia Nielsen   45          Denmark
9   Lucas Miller   53           Canada
```

In [ ]: